

UNIVERSITY OF CANTERBURY

EMTH210

ENGINEERING MATHEMATICS

Topic 3: Fourier series

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3.1 FOURIER SERIES

A 2π -periodic function $f(x)$ can have a Fourier Series (F.S.) associated with it:

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos(nx) + b_n \sin(nx) \right),$$

where the coefficients $a_0, \{a_n\}_{n=1}^{\infty}, \{b_n\}_{n=1}^{\infty}$ depend in some way on $f(x)$ (for a preview, check out the box on page 4). The important point is that for a “nice” function, its associated Fourier series is actually equal to the function itself, so we can write:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos(nx) + b_n \sin(nx) \right).$$

When we encounter an inhomogeneous ODE with a RHS which we can’t tackle by the method of undetermined coefficients, we will use the FS to rewrite the RHS in terms of sines and cosines, which *can* be tackled with the method of undetermined coefficients. There are many other uses of FS, which in turn is just one part of the wider subject of Fourier analysis. Joseph Fourier¹ (1768–1830) invented FS in order to solve the heat equation in a metal plate.

It might be that only a finite number of the coefficients a_n and b_n are non-zero, in which case we have a finite Fourier series. On the other hand, it may be that an infinite number of the coefficients are non-zero.

Note that because $\sin(nx)$ and $\cos(nx)$ are 2π -periodic, (or rather 2π is the shortest period common to all), the FS is also 2π periodic.

¹Interesting chap. An important part of the French Revolution, later chummy with Napoleon Bonaparte. Discovered the greenhouse effect. Was keen on heat flow (Fourier’s law) and vibrations (Fourier analysis) in general: ironically, tripped on the blanket he was wearing to keep warm and died after falling down the stairs.

An example of a Fourier series is that if $f(x) = x$ on $-\pi < x < \pi$ then

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(nx),$$

ie $a_0 = \underline{\hspace{2cm}}$, $a_n = \underline{\hspace{2cm}}$ for all n , and $b_n = \underline{\hspace{2cm}}$ for all n .

For the idea of Fourier series to be useful, we need to know how to get a_n and b_n . We do so by use of the *orthogonality property* of the set $\mathcal{S}$ of functions

$$\mathcal{S} = \{1, \sin(x), \cos(x), \sin(2x), \cos(2x), \dots\},$$

which is that

for any two *different* functions $\phi_1(x)$ and $\phi_2(x)$ in $\mathcal{S}$.

This can be a very useful shortcut.

Example 3.1.1

Show that the orthogonality property holds for the pair of functions $\phi_1(x) = \sin(nx)$, $\phi_2(x) = \sin(kx)$ where k and n are natural numbers, and $k \neq n$.

Solution.



Now, when $\phi_1(x) = \phi_2(x)$ we get either

$$\begin{aligned}\int_{-\pi}^{\pi} \sin^2(kx) \, dx &= \pi \quad \text{or} \\ \int_{-\pi}^{\pi} \cos^2(kx) \, dx &= \pi \quad \text{or} \\ \int_{-\pi}^{\pi} 1^2 \, dx &= 2\pi.\end{aligned}$$

We use orthogonality and these last three properties to get the FS coefficients.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)),$$

where

$$\begin{aligned}a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) \, dx \\ b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) \, dx.\end{aligned}$$

Note that some books omit the $1/2$ from the $a_0/2$ term, in which case the factor in front of the a_0 integral becomes $1/2\pi$. Either way, you get the same series, of course. But it is best to stick with one definition, and we will use the one in the box.

The above concerns functions on the symmetric interval $(-\pi, \pi)$. Later, we will generalise to any interval, at which point our definition of the FS and its coefficients will change. The general version is the one on the formula sheet.

Example 3.1.2

Derive the formula for b_n by considering $\int_{-\pi}^{\pi} f(x) \sin(nx) \, dx$.

Solution.



The other derivations are similar.

3.2 ODD AND EVEN FUNCTIONS

A function is *odd* if it satisfies $f(-x) = -f(x)$. E.g. $x, x^3, x^5, x^7, \sin x, \tan x, \sinh x$.

A function is *even* if it satisfies $f(-x) = f(x)$. E.g. $1, x^2, x^4, x^6, \cos x, \cosh x, |x|$.

They satisfy the following multiplication and addition properties.

$$\begin{aligned} \text{odd} \times \text{odd} &= \text{even}, & \text{odd} \times \text{even} &= \text{odd}, & \text{and} & & \text{even} \times \text{even} &= \text{even} \\ \text{odd} + \text{odd} &= \text{odd}, & \text{odd} + \text{even} &= \text{neither}, & \text{and} & & \text{even} + \text{even} &= \text{even} \end{aligned}$$

ODD FUNCTIONS

For an odd function $f(x)$,

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx = 0,$$

since the interval is symmetric about 0. $f(x) \cos(nx)$ is odd, so

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) \, dx = 0,$$

for all n , but $f(x) \sin(nx)$ is even, so

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx) \, dx,$$

for all n .

EVEN FUNCTIONS

For an even function $f(x)$,

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_0^{\pi} f(x) \, dx \\ a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx) \, dx, \end{aligned}$$

for all n , as $f(x) \cos(nx)$ is even, but $f(x) \sin(nx)$ is odd so

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) \, dx = 0$$

for all n .

Example 3.2.1

Find the Fourier series of

$$f(x) = x \quad \text{on} \quad -\pi < x < \pi,$$

with $f(x + 2\pi) = f(x)$.

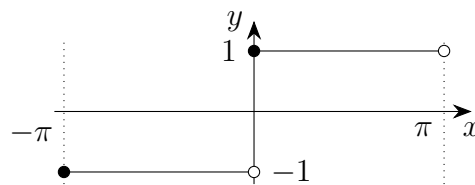
Solution.



Example 3.2.2

Find the Fourier series of the following function:

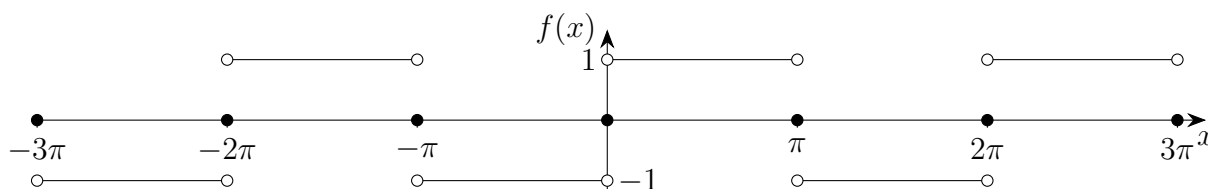
$$f(x) = \begin{cases} -1 & \text{if } -\pi \leq x < 0 \\ 1 & \text{if } 0 \leq x < \pi, \end{cases} \quad \text{with } f(x + 2\pi) = f(x).$$



Solution.



If we were also asked to sketch the FS, we would draw something like this:

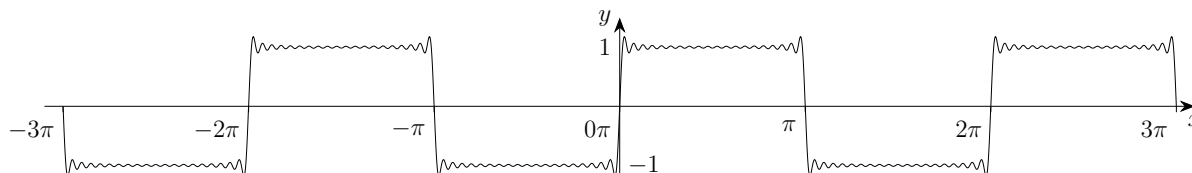


3.3 GIBBS'S PHENOMENON

We have already seen that the function

$$f(x) = \begin{cases} -1 & -\pi \leq x < 0 \\ 1 & 0 \leq x < \pi \end{cases} \quad \text{has FS} \quad f(x) = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{4}{n\pi} \sin(nx).$$

If we include only the first N terms as an approximation to $f(x)$ we get erroneous oscillations close to corners of the square wave. Here are the first 20 terms:



If the file is played as sound (or if the phenomenon is a naturally aural one) then these oscillations make a ringing sound. This phenomenon is named not after its discoverer (typical), but rather its populariser, JW Gibbs. The maximum height of the oscillations near the jump discontinuities do not go to zero as $N \rightarrow \infty$. This causes all manner of problems in signal processing, for example in MRI scans.

3.4 FUNDAMENTAL HARMONICS

Consider the following finite FS:

$$f(x) = 4 + \sin(x) + 12 \sin(2x) + 35 \sin(5x).$$

The ‘4’ is the constant term which is the average value of $f(x)$ over one period. The shortest period common to all terms is 2π . The *fundamental frequency* is $1/(\text{shortest period})$. Here,

$$\text{fundamental frequency} = \frac{1}{2\pi}.$$

- The $\sin(x)$ term is the *fundamental frequency term*², because its frequency (the inverse of its period) is equal to the fundamental frequency.
- The $12 \sin(2x)$ term is the *2nd harmonic*³, because its frequency is twice the fundamental frequency (so not just because there is a ‘2’ in there).
- The $35 \sin(5x)$ term is the *5th harmonic*⁴, because its frequency is five times the fundamental frequency.

In the above example, the FS is missing some harmonics, namely the _____, _____, and everything above the _____. Likewise, a FS may also miss a fundamental term, although it will always have a fundamental frequency, as in the following example.

²Musicians among you may know this as the *fundamental tone*.

³Musicians: *1st overtone*.

⁴Musicians: *4th overtone*.

Example 3.4.1

Find the fundamental frequency and identify the harmonic terms of the FS

$$f(x) = 4 + \sin(2x) + \cos(3x).$$

Solution.



3.5 ANOTHER EXAMPLE

Example 3.5.1

Find the Fourier series of the function

$$f(x) = \pi - |x| \quad -\pi < x < \pi,$$

with $f(x + 2\pi) = f(x)$.

Solution.



3.6 NOT A FOURIER SERIES

Example 3.6.1

Why is the following not a FS?

$$f(x) = \cos(x) + \frac{1}{2} \cos(\sqrt{2}x) + \frac{1}{4} \cos(2x) + \frac{1}{8} \cos(\sqrt{8}x) + \dots$$

Solution.

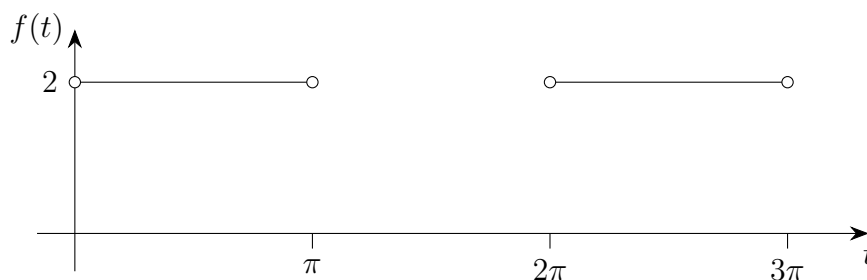


3.7 F.S. IN UNDETERMINED COEFFICIENTS

The main use of FS in this course will be to rewrite unusual right hand sides of ODEs in terms of sines and cosines, which we know how to deal with in the method of undetermined coefficients. Here is an illustrative example.

Example 3.7.1

A mass $m = 1\text{kg}$ is suspended on a spring which hangs vertically. The spring obeys Hooke's law with a spring constant k whose square root is not a natural number. Let t be time and y be the displacement of the mass from the equilibrium position, where downwards is positive, and is the direction in which gravity acts. The mass-spring system is subject to forcing from the force $f(t)$. Find the position of the mass at all positive times when the forcing takes the form



and continues in this way for all positive times.

Solution.



3.8 GENERALISATION TO ARBITRARY PERIOD p

The following is very important, because we generalise what we have learned about FS of 2π -periodic functions to enable us to find the FS of a p -periodic function⁵.

So let us take a 2π -periodic function which we have already studied, namely

$$f(x) = x \quad \text{on} \quad -\pi < x < \pi \quad \text{with} \quad f(x) = f(x + 2\pi).$$

which has Fourier series $f(x) = \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(nx)$.

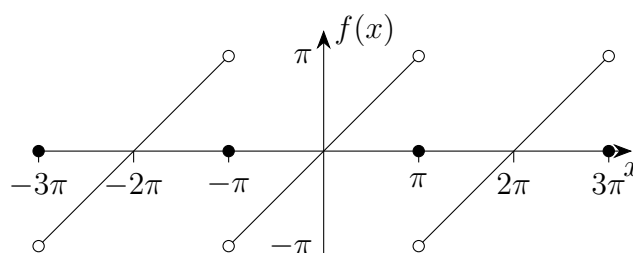
Now suppose that in both the original definition and the FS, we put $x = 2\pi t/p$, where p is some non-zero number, and t is a variable:

$$f(x) = f\left(\frac{2\pi t}{p}\right) = g(t) \quad \text{say.}$$

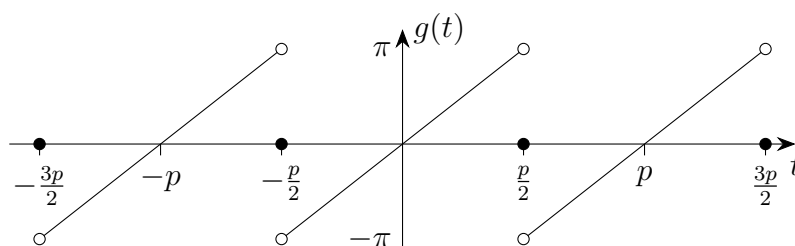
Now our new function $g(t)$ has period p . The Fourier series for $g(t)$ is simply

$$g(t) = \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin\left(\frac{2n\pi t}{p}\right),$$

after making the substitution. Graphically, we have gone from



to



⁵What we don't get into here, but which is nevertheless important, is that almost any function can be thought of as p -periodic for some p , if you truncate it, focus on only a certain range of interest, or chop it into suitable pieces.

We can generalise the definition of Fourier series to work with any periodic function $f(x)$ of period p :

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \left(\frac{2n\pi x}{p} \right) + b_n \sin \left(\frac{2n\pi x}{p} \right) \right)$$

where

$$\begin{aligned} a_0 &= \frac{2}{p} \int_{\text{one period}} f(x) \, dx \\ a_n &= \frac{2}{p} \int_{\text{one period}} f(x) \cos \left(\frac{2n\pi x}{p} \right) \, dx \\ b_n &= \frac{2}{p} \int_{\text{one period}} f(x) \sin \left(\frac{2n\pi x}{p} \right) \, dx \end{aligned}$$

Note that we do not have to integrate from $-p/2$ to $p/2$. It is sufficient to integrate over any interval $[A, A + p]$, for some convenient A .

The above can be proved by orthogonality on the set

$$\left\{ 1, \sin \left(\frac{2n\pi x}{p} \right), \cos \left(\frac{2n\pi x}{p} \right), \dots \right\}.$$

If we set $p = 2\pi$ then we retrieve exactly the definition we have been working with. You are much more likely to be tested on the general form of the FS than the FS of a 2π -periodic function. Also, it is this general form which appears on the formula sheet.

Example 3.8.1

Find the FS of the function

$$f(x) = \begin{cases} 0 & -1 \leq x < 0 \\ \sin(\pi x) & 0 \leq x < 1. \end{cases}$$

Solution.

